A web server for enzyme function prediction using active and binding amino acids

Gopal Peddinti[1], Chunhao Gu[2], Samuli Ollila[1], Paula Jouhten[2] and Sandra Castillo[1]

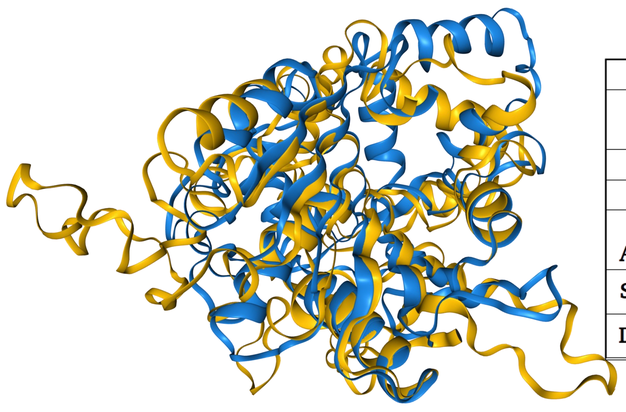
[1]VTT Technical Research Center of Finland, Espoo, Finland. [2]Aalto University, Espoo, Finland

Aim of enzyme function prediction based on protein structure

EC number	1.1.1.2
Name	alcohol dehydrogenase (NADP(+))
Alternative name	aldehyde reductase (NADPH)
Reaction	a primary alcohol + NADP(+) = an aldehyde + NADPH + H(+)

Challenges in enzyme function prediction

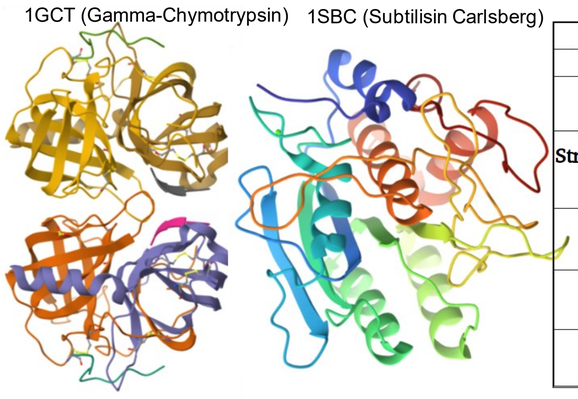
Similar structures may have different functions



Structural alignment of Uniprot proteins Q9ZHI2 and Q07950

Uniprot ID: Q07950	Uniprot ID: Q9ZHI2
Sterol esterase 2	Poly(3-hydroxyalkanoate) polymerase
Saccharomyces cerevisiae	Chromobacterium violaceum
EC number: 3.1.1.3	EC number: 2.3.1.140
Active site with Serine-Aspartic acid-Histidine triad	Active site with Cysteine
Significant structural alignment (i.e., similar overall structure)	
Different enzyme function due to differences in the active site	

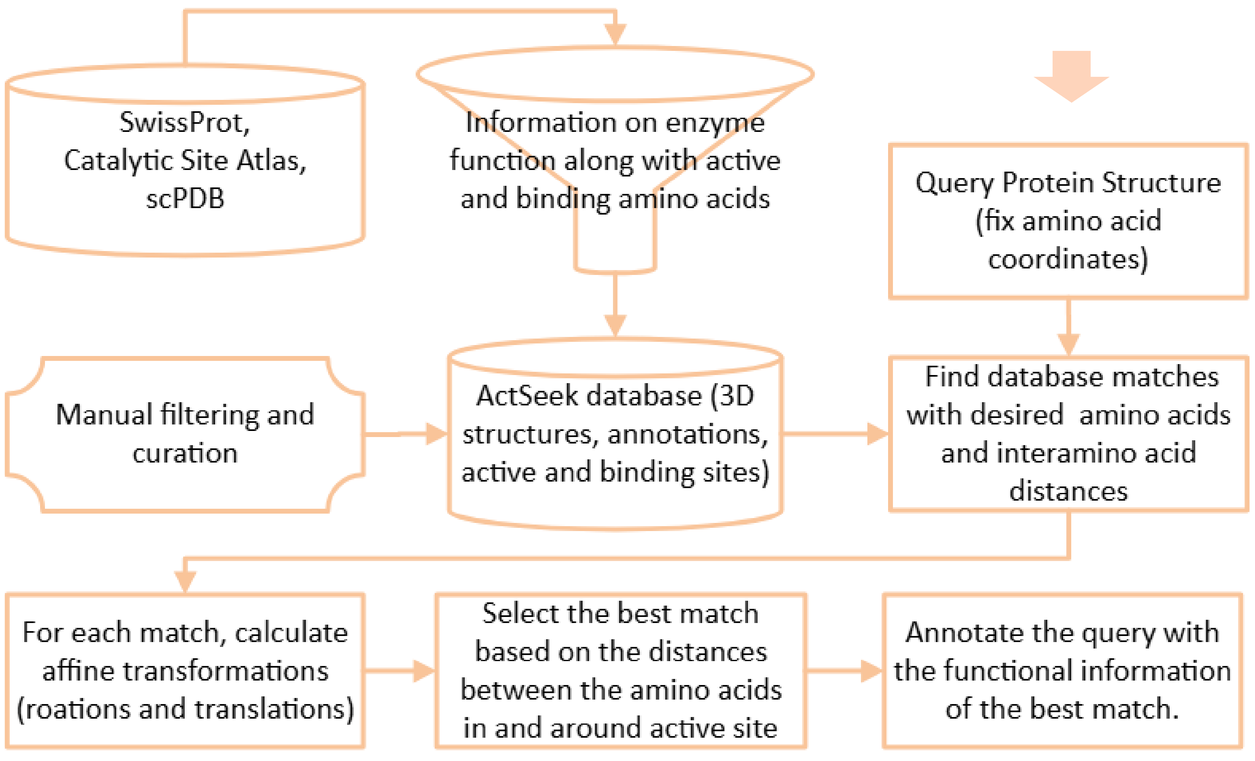
Dissimilar structures may have similar functions



1GCT	1SBC
EC: 3.4.21.1	EC: 3.4.21.64
Peptidase family S1	Peptidase family S8
Structure: composed of two β-barrel domains	Structure: single-domain α/β fold.
Use the same catalytic triad (Serine-Histidine-Aspartate)	
Cleave peptide bonds with similar specificities	
Distinct substrate specificities and mechanisms of action	

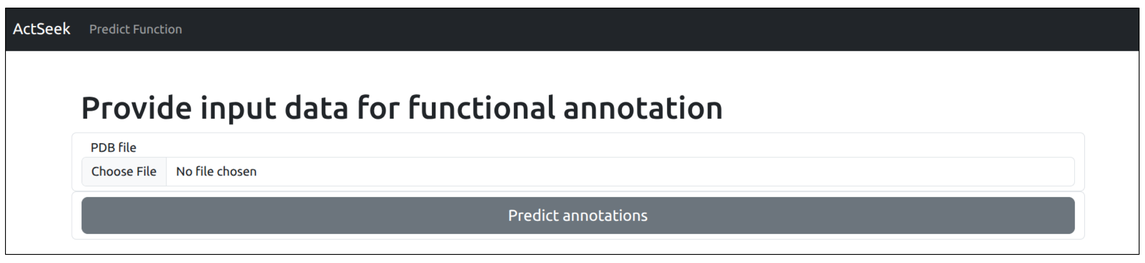
Solution provided by ActSeekAnn

ActSeekAnn predicts annotations based on the spatial coordinates of specific amino acids (active or binding) crucial for enzyme functionality. We have collected and curated a database of AlphaFold structures and enzyme function annotations along with the active and binding sites related to the enzyme function. ActSeek algorithm finds the hits based on the distance between the active and binding amino acids. To reduce false positives, it also considers a few amino acids in the neighbourhood of the active site.

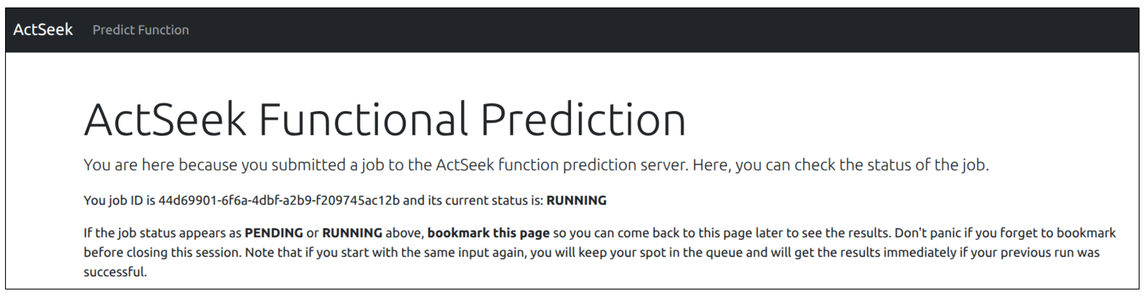


ActSeekAnn web server

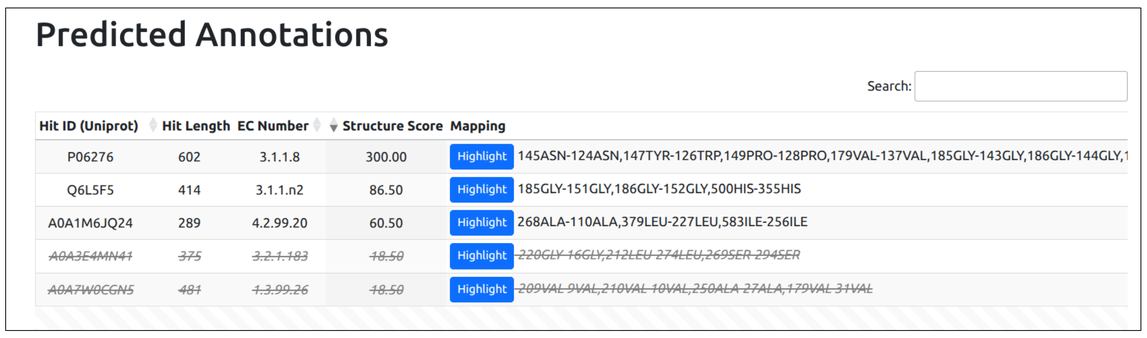
To make the ActSeek-based functional annotations accessible to the public, we have developed a Django-based web server, available at the URL: **https://actseek.vtt.fi**. The user can upload a PDB file to describe the 3D structure of the query protein. The web server processes the requests serially on a first-in-first-out basis, while queuing other requests for processing.



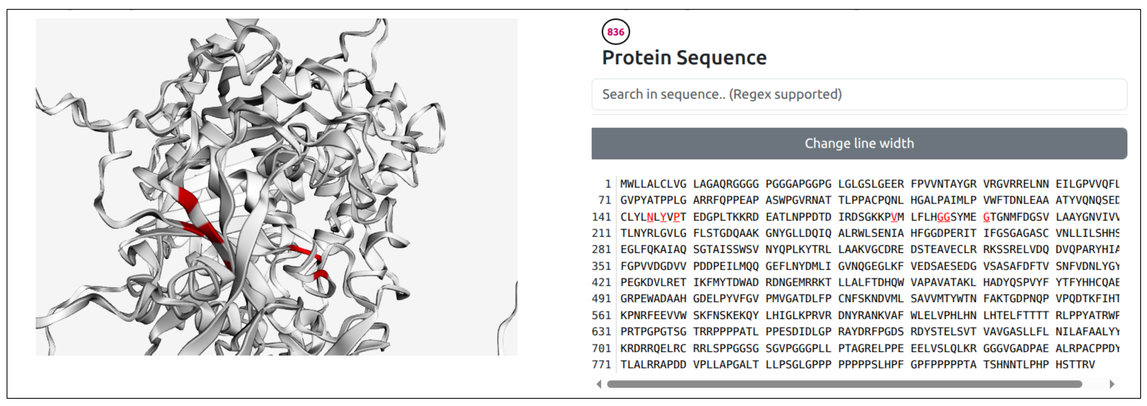
On the submission form, the user can upload a PDB file to describe the 3D structure of the query protein. The web server processes the requests serially on a first-in-first-out basis, while queuing other requests for processing.



The function prediction request is immediately added to the job queue. If no other prediction jobs running or waiting in the queue, the job starts running, as shown in the status page above. If the job is still waiting to be run because there are other jobs currently running or in the queue ahead of the recently submitted job, the status appears as PENDING. The page can be bookmarked to check the status of the job later. The web browser window may be closed at this stage while the job runs on the server.



The results page of a successfully run job contains a table providing the predicted EC number, the Uniprot ID of the hit whose active sites align well with query protein, and some statistics of the prediction. Potentially unreliable hits (score ≤ 20) are clearly marked in the table. The highlight button appearing next to the mapping of the active and binding amino acids allows the user to visualize the amino acids in the query protein's structure and sequence as shown below.



Conclusions

- ActseekAnn is a precise method integrated with the AlphaFold database, enabling the annotation of enzymes based on the coordinates of a few amino acids (partial structural homology).
- The quality of these annotations depends on the quality of the annotated proteins in SwissProt.
- We developed a web server to make ActSeek-based functional annotations accessible to the community.

References

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[2] Varadi, M et al. AlphaFold Protein Structure Database: massively expanding the structural coverage of protein-sequence space with high-accuracy models. Nucleic Acids Research (2022).

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[4] Furnham, N, Holliday, GL, de Beer, TAP, et al., The Catalytic Site Atlas 2.0: cataloging catalytic sites and residues identified in enzymes, NAR, 42, D1, D485–D489 (2014).

[5] Desaphy J, Bret G, Rognan D, Kellenberger E. sc-PDB: a 3D-database of ligandable binding sites--10 years on. NAR, 43(Database issue):D399-404, (2015).

ActSeekAnn Web Server URL: **https://actseek.vtt.fi**

Contact: Gopal Peddinti, Senior Scientist
Tel. +358 401427562, gopal.peddinti@vtt.fi

www.vttresearch.com

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